TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

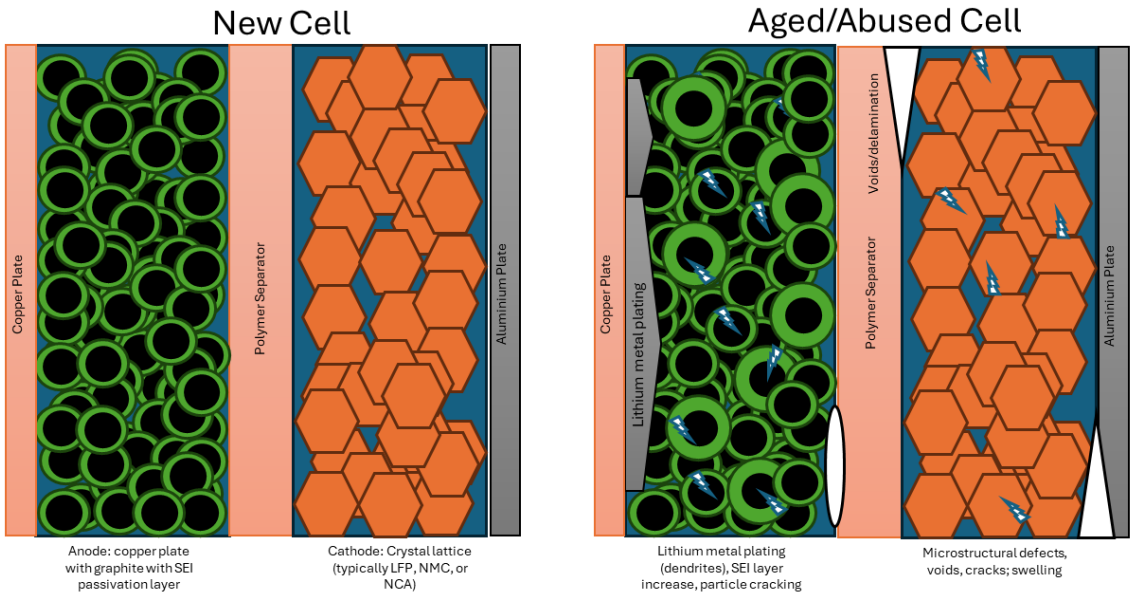
The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Subhajeet Rath	Email	subhajeet.rath@tno.nl
Institute Name	Nederlandse Organisatie voor Toegepast Natuurwetenschappelijk Onderzoek (TNO), Powertrains Department			
Address	Automotive Campus 30, 5708 JZ Helmond, The Netherlands			

2. Technology Overview

Title of Proposed Technology	Electrochemical Impedance Spectroscopy (EIS) and AI-based non-destructive battery safety and health diagnostics for EV and ESS (cell, module, and pack).			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (V), Advanced Mobility ( ), Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cyber Security ( ), Artificial Intelligence (V), Next-generation Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Instrumentation	Safety monitoring/diagnosis measurement and control instruments	Batteries	Secondary Batteries
Concept of Proposed Technology	TNO proposes a joint R&D project (TIPA 2026) with a Korean SME partner to develop and validate a non-destructive battery safety and health diagnostic solution for electric vehicle (EV) and energy storage system (ESS) batteries. This solution is achieved by combining the following elements: • Electrochemical Impedance Spectroscopy (EIS) measurements, utilizing measurement hardware and workflow integration provided by the Korean SME partner • Physics-based battery models and Battery Management System (BMS) algorithms,			

	complemented by data-driven anomaly detection and state estimation techniques developed by TNO The project focuses on robust detection of abnormal states and early safety risks, as well as scalable state-of-X (SoX) estimation – including state-of-charge (SoC), state-of-health (SoH), and state-of-power (SoP) – across a wide range of battery types, temperatures, and usage profiles. These capabilities will be verified through controlled laboratory testing and field pilots to ensure effectiveness under real-world conditions.
Rationale for Joint R&D Project	The rapid scale-up of EV and ESS deployment is significantly increasing the demand for non-destructive, fast, and reliable battery safety and health diagnostics, especially to inform inspection, maintenance, reuse, and recycling decisions. As battery packs age, safety risks and performance degradation become increasingly heterogeneous across cells, while destructive testing remains costly and impractical for large-scale screening. Recent research indicates that Electrochemical Impedance Spectroscopy (EIS), when combined with physics-based modeling and AI-driven analysis, is a promising approach to detect abnormal states, internal degradation, and incipient safety risks under various operating conditions [1][2][5][6].  The Korean SME partner contributes industrial EIS measurement hardware, established battery inspection workflows, and commercialization pathways. However, this partner requires advanced algorithm development and independent validation to ensure diagnostic robustness across different cell chemistries, battery pack designs, and operating environments. TNO brings complementary, internationally validated expertise in battery modeling, BMS algorithms, EIS data interpretation, and AI-based anomaly detection, supported by controlled testing infrastructure from cell to pack level (including climate chambers and Model-in-the-Loop / Hardware-in-the-Loop (MIL/HIL) verification) [2][3][6]. The joint R&D project combines deployment-driven industrial needs with research-grade validation and algorithm development,

	thereby reducing technical risk and accelerating the Technology Readiness Level (TRL) advancement toward export-ready diagnostic solutions that align with Korea's strategic priorities in Secondary Batteries and Artificial Intelligence [3][5].
Key Words	1. EIS 2. Battery diagnostics 3. Non-destructive inspection 4. AI anomaly detection 5. EV and ESS safety

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	(5)	End of the Project	(8)
R&D Objectives	The objective of this joint R&D project is to advance non-destructive battery safety and health diagnostic technologies for electric vehicle (EV) and energy storage system (ESS) applications by combining electrochemical impedance spectroscopy (EIS), physics-based battery modeling, and data-driven analytics. The R&D objectives are defined as: • Development of robust methodologies for early detection of abnormal battery states and emerging safety risks using impedance-based diagnostics • Advancement of scalable, non-destructive state estimation methods for battery state-of-charge (SoC), state-of-health (SoH), and state-of-power (SoP) across battery chemistries, formats, and operating conditions • Improvement of diagnostic robustness and transferability across temperature ranges, aging states, and usage profiles • Establishment of validated, explainable diagnostic outputs suitable for inspection, maintenance, reuse, and recycling decision-making • Demonstration of technology readiness through laboratory validation and representative field-relevant use cases			
R&D Target Specifications	The following target specifications reflect the generic R&D objectives and are intended as performance goals to be validated through controlled testing and pilot demonstrations. Final values will be refined during the early phase of the project based on test results and system constraints. Diagnostic time: • Target: ≤ 20 minutes per battery pack for non-destructive screening, including EIS measurement and analysis SoH estimation accuracy: • Target: ≤ 7% absolute error relative to reference capacity and resistance-based measurements under controlled conditions Abnormality and safety-risk detection: • High sensitivity to impedance signatures associated with degradation and abnormal behavior • False-alarm rate suitable for operational screening scenarios			

	• Diagnostic outputs providing interpretable indicators (e.g., degradation mode likelihoods or confidence levels) Operational robustness: • Validated performance across a representative range of temperatures, states of charge, and aging conditions • Applicability to both EV and ESS battery modules and packs Interoperability and scalability: • Diagnostic methods compatible with integration into inspection systems, data platforms, or BMS-level workflows • Scalability from module-level to pack-level assessment
R&D Plan	Draft R&D Plan (36 Month indicative) **WP1 – Requirements Definition and Diagnostic Framework Design (Month 1–6)** • Definition of functional and performance requirements for non-destructive battery diagnostics • Identification of relevant safety, degradation, and abnormality scenarios for EV and ESS batteries • Definition of data structures, interfaces, and evaluation metrics for EIS-based diagnostics • Design of the overall diagnostic framework combining EIS measurements, physics-based models, and data-driven analysis Deliverables: • System requirements specification • Diagnostic framework and evaluation methodology **WP2 – Data Acquisition and Experimental Validation Setup (Month 4–14)** • Design of experimental test protocols at cell, module, and pack levels • Acquisition of EIS and reference measurement datasets under controlled laboratory conditions • Execution of aging, stress, and fault-representative test scenarios • Establishment of reference datasets for model calibration and validation Deliverables: • Validated experimental protocols • Curated and documented diagnostic datasets **WP3 – Model-Based and Data-Driven Diagnostic Algorithm Development (Month 10–26)**

	• Development of physics-based impedance and battery behavior models • Feature extraction and representation of EIS signatures for diagnostic purposes • Development of data-driven methods for anomaly detection and SoX estimation • Integration of uncertainty quantification and explainability concepts into diagnostic outputs Deliverables: • Prototype diagnostic algorithms • Performance assessment reports against defined metrics **WP4 – System Integration and Performance Evaluation (Month 20–32)** • Integration of diagnostic algorithms into a representative diagnostic system environment • Validation of diagnostic performance under realistic operating conditions • Robustness assessment across battery types, temperatures, and aging states • Iterative refinement based on validation results Deliverables: • Integrated diagnostic prototype • Validation and robustness evaluation report **WP5 – Pilot Demonstration, Documentation, and TRL Advancement (Month 30–36)** • Pilot-scale demonstration using representative EV and ESS battery systems • Assessment of technology readiness against TRL 8 criteria • Preparation of technical documentation supporting deployment and further commercialization • Dissemination activities, including technical reports and publications where applicable Deliverables: • Pilot demonstration results • Final project report and TRL assessment

4. R&R (Roles and Responsibilities)

Expected R&R of Proposer (TNO)	TNO (as a research institute) is expected to: • Provide extensive battery modeling and BMS algorithm development capabilities, including SoX estimation, parameter identification, balancing and control strategies, thermal state estimation, and battery aging modeling [7]. • Support rigorous development workflows (e.g. model-based design) and verification methodologies (e.g. MIL/HIL simulation) to ensure the diagnostic algorithms are robust and ready for BMS integration. • Design and execute controlled testing and validation campaigns at cell, module, and pack levels (utilizing climate chambers and electrical test benches) to quantify diagnostic performance and robustness [4][8]. • Contribute to dissemination of results via publications and produce standardization-relevant outputs (for example, developing traceability concepts for a battery "passport" data system).
Expected R&R of Korean SME Partner	The Korean SME partner is expected to: • Provide the EIS measurement equipment and integrate the diagnostic solution into its EV/ESS battery inspection workflow. • Supply the operational context, baseline software platform, and relevant data assets needed for joint algorithm development. • Implement the jointly developed algorithms into the company's products, lead pilot deployments in Korea, and drive subsequent commercialization efforts.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes of the joint R&D project include: • Improved safety and health assessment for EV and ESS batteries through non-destructive, impedance-based diagnostic techniques. • Faster and lower-cost decision support for battery maintenance, resale, second-life reuse, and recycling workflows. • Demonstrated performance and reliability of the diagnostics via joint laboratory and field validation, building confidence for global market deployment. • Generation of potential intellectual property (IP) and research publications, as well as contributions to interoperability and standardization efforts in battery diagnostics and safety assessment.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization